

# 1.8 LONGITUDINAL & TRANSVERSE MAGNETIZATION

RF excitation tips  $M_z$  into the transverse plane to create MR signal.

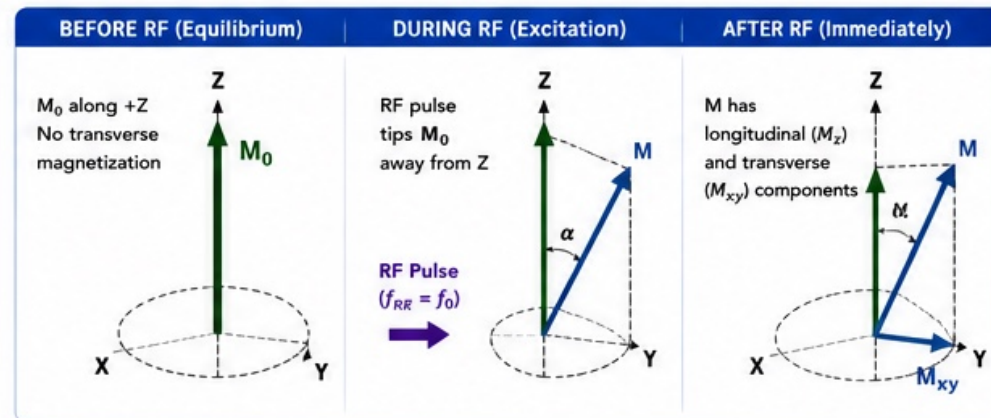
At equilibrium, net magnetization ( $M_0$ ) is along +Z (longitudinal).

An RF pulse at the Larmor frequency tips  $M_0$  away from Z.

The transverse component ( $M_{xy}$ ) circles the Z-axis (precesses)

and induces voltage in the receive coil.

Only  $M_{xy}$  produces the measurable MR signal.



## COMPONENTS AFTER EXCITATION

For a flip angle  $\alpha$ :

$$M_z = M_0 \cos \alpha$$

$$M_{xy} = M_0 \sin \alpha$$

$\alpha$  = flip angle (degrees)

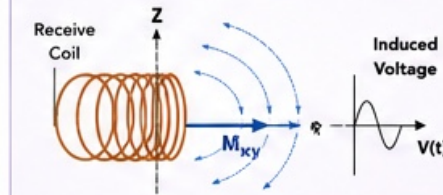
$M_z$  = longitudinal magnetization

$M_{xy}$  = transverse magnetization

$M_0$  = equilibrium magnetization

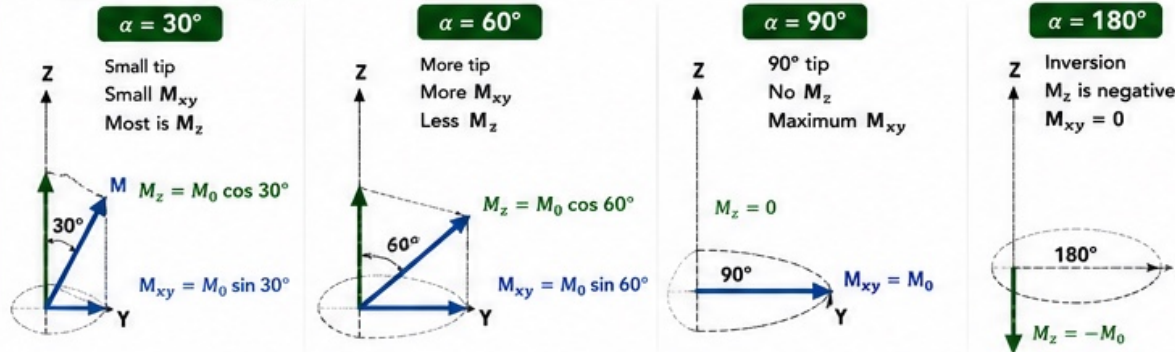
## WHY ONLY TRANSVERSE MAGNETIZATION PRODUCES SIGNAL

Precessing  $M_{xy}$  creates a changing magnetic field through the receive coil  $\rightarrow$  induces voltage (Faraday's Law).



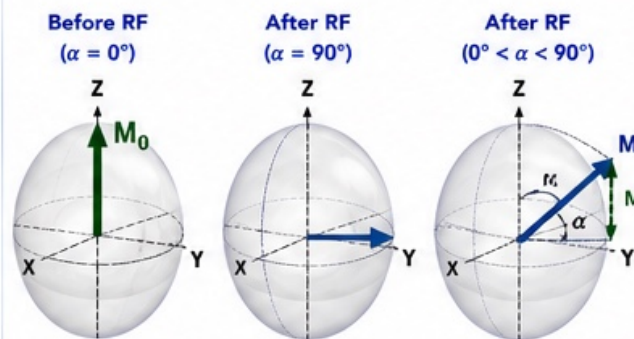
$M_z$  is parallel to  $B_0 \rightarrow$  no changing flux  $\rightarrow$  no voltage  $\rightarrow$  no signal.

## EFFECT OF FLIP ANGLE ( $\alpha$ )



Larger flip angle  $\rightarrow$  more transverse magnetization  $\rightarrow$  stronger signal (up to 90°). Beyond 90°,  $M_z$  becomes negative (inversion).

## 3D VIEW (BLOCH SPHERE REPRESENTATION)



## IMPORTANT POINTS

- RF pulse does not create magnetization. It only changes the direction.
- After RF:  $M$  has both  $M_z$  and  $M_{xy}$ .
- Transverse magnetization ( $M_{xy}$ ) precesses at  $f_0$  and produces signal.
- After the RF pulse ends,  $M_z$  and  $M_{xy}$  return toward equilibrium (relaxation).

## ENGINEERING INSIGHT

- The receive system is blind to  $M_z$ . No  $M_{xy} \rightarrow$  No signal.
- Signal strength is proportional to the amount of  $M_{xy}$  and the efficiency of the receive coil.
- Flip angle errors directly reduce  $M_{xy}$  and image SNR.

Large  $M_{xy}$   
(strong signal)



Small  $M_{xy}$   
(weak signal)



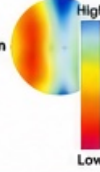
## FIELD SERVICE TIP

- $B_1$  field uniformity determines the actual flip angle.
- Check flip angle calibration during PM.
- Inhomogeneous  $B_1 \rightarrow$  spatial variations in flip angle  $\rightarrow$  uneven signal and contrast.

Good  $B_1$   
Uniform flip angle



Poor  $B_1$   
Non-uniform flip angle



## COMMON MISCONCEPTION

"More RF power always gives a better image."

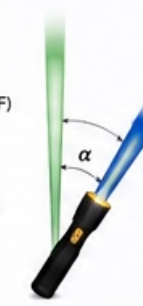
**Reality:** RF power beyond 90° does not increase signal. In fact, it reduces  $M_z$  and wastes power. Proper flip angle is essential for optimal contrast and SNR.

## REAL-WORLD ANALOGY

Imagine a flashlight ( $M_0$ ) shining straight up.

Tilt it toward the horizon (apply RF)  $\rightarrow$  now part of the light shines horizontally ( $M_{xy}$ ).

We can only "see" the horizontal light, not the vertical.



## KEY TAKEAWAY

RF excitation tips  $M_0$  away from +Z.

This creates transverse magnetization ( $M_{xy}$ ) which precesses and generates the MR signal.

The amount of  $M_{xy}$  depends on the flip angle.

Only transverse magnetization is detectable.

